

LACOMBE, AB, Canada

WMO: 712420

Lat: 52.4492N

Lon: 113.7569W

Elev: 2822

StdP: 13.26

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-18.5	-30.5	1.1	-24.8	-24.0	1.6	-18.2	23.0	24.5	20.1	18.6	3.2	170	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	82.6	64.5	79.2	62.9	76.1	61.3	67.2	78.8	65.0	76.2	63.0	73.5	7.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	95.4	73.7	60.5	87.5	71.1	58.4	81.1	68.7	33.4	79.2	31.5	76.4	29.9	73.7	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	17.9	15.3	-33.3	88.6	7.1	3.4	-38.4	91.1	-42.6	93.1	-46.5	95.0	-51.7	97.5
			-33.4	71.1	6.8	2.7	-38.3	73.0	-42.2	74.5	-46.0	76.0	-51.0	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	12.0	14.7	24.3	38.4	49.6	56.7	61.3	59.2	50.7	39.0	24.1	13.6	
	DBStd	20.69	15.49	13.72	13.85	9.24	7.95	5.45	5.25	5.67	7.55	8.33	12.28	13.59	
	HDD50	5771	1179	987	796	356	105	8	1	4	80	350	777	1128	
	HDD65	10225	1644	1407	1260	797	478	253	139	192	429	806	1227	1593	
	CDD50	1063	0	0	0	8	93	210	351	289	102	9	1	0	
	CDD65	43	0	0	0	0	1	5	24	12	1	0	0	0	
	CDH74	1144	0	0	0	3	89	129	431	361	125	6	0	0	
	CDH80	227	0	0	0	0	12	16	96	78	25	0	0	0	
Wind	WSAvg	6.2	5.4	5.3	6.1	7.6	7.8	7.2	6.3	5.6	6.0	6.5	5.5	5.3	
Precipitation	PrecAvg	17.9	0.8	0.7	0.6	0.9	1.9	3.0	3.5	2.7	2.0	0.7	0.6	0.7	
	PrecMax	22.3	2.6	1.7	1.6	2.2	5.4	7.3	7.3	6.6	4.6	1.9	1.4	2.1	
	PrecMin	12.1	0.0	0.0	0.0	0.0	0.4	1.0	0.6	0.4	0.1	0.0	0.0	0.0	
	PrecStd	2.8	0.6	0.4	0.4	0.6	1.1	1.6	1.6	1.5	1.2	0.5	0.4	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	45.8	56.1	72.4	81.6	82.0	87.0	87.1	83.7	72.4	57.2	45.5	
		MCWB	37.8	37.8	43.5	52.0	57.4	63.2	68.0	65.2	61.4	53.6	46.2	37.8	
	2%	DB	41.2	41.2	49.8	66.6	76.2	77.8	82.7	81.6	77.6	64.3	49.0	40.5	
		MCWB	35.5	34.8	40.3	49.5	55.6	60.8	66.9	64.3	59.8	50.8	40.7	34.7	
	5%	DB	36.8	37.4	45.7	60.5	71.6	74.1	78.8	78.0	72.6	58.6	43.3	36.2	
		MCWB	32.6	32.2	38.2	45.7	53.8	59.0	65.2	63.3	57.4	47.2	37.1	31.6	
	10%	DB	32.5	33.4	41.7	55.0	66.9	70.4	75.2	74.3	67.4	53.7	39.5	32.1	
		MCWB	29.2	29.6	35.6	42.9	51.5	57.2	63.2	62.1	55.1	44.5	34.6	28.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.4	44.8	52.9	59.5	66.1	71.5	68.9	63.4	55.1	46.5	38.7
			MCDB	44.3	44.9	55.0	70.4	76.7	78.2	81.0	80.0	79.4	69.5	56.5	44.8
2%		WB	35.7	35.2	40.9	49.8	56.7	62.7	68.3	66.4	60.3	51.2	41.2	34.9	
		MCDB	40.7	40.8	48.8	65.5	72.9	74.0	79.7	78.1	74.7	63.1	47.9	39.7	
5%		WB	32.9	32.4	38.3	46.6	54.7	60.3	66.0	64.4	58.0	48.0	37.8	31.8	
		MCDB	36.7	36.8	45.0	59.0	69.4	71.0	77.2	76.0	71.4	57.2	42.6	35.9	
10%		WB	29.3	29.7	35.8	43.6	52.5	58.3	63.9	62.3	55.5	45.0	34.7	28.8	
		MCDB	32.5	33.2	41.3	53.6	65.3	68.6	73.9	73.1	66.3	52.6	38.7	32.1	
Mean Daily Temperature Range	5% DB	MDBR	20.1	22.7	21.4	22.6	24.1	21.6	23.1	24.8	25.1	23.0	19.1	19.3	
		MCDBR	23.4	26.4	26.1	33.8	33.0	29.2	29.8	33.0	36.3	32.1	24.7	21.6	
		MCWBR	18.5	20.4	17.6	18.6	15.8	14.3	15.9	17.6	19.4	19.4	17.1	16.7	
	5% WB	MCDBR	23.0	25.5	25.0	31.7	29.8	25.4	27.7	29.8	33.9	31.1	23.6	21.3	
		MCWBR	18.3	19.8	17.2	18.1	14.5	13.4	15.5	16.8	18.5	19.1	16.8	16.8	
Clear-Sky Solar Irradiance	taub	0.261	0.266	0.294	0.325	0.337	0.344	0.346	0.358	0.309	0.283	0.254	0.244		
	taud	2.312	2.390	2.409	2.376	2.399	2.402	2.422	2.362	2.482	2.536	2.470	2.319		
	Ebn at Noon	237	275	287	289	290	288	285	275	279	267	245	224		
	Edh at Noon	19	24	29	34	35	35	34	34	27	21	16	16		
All-Sky Solar Radiation	RadAvg	337	659	1103	1486	1796	1900	1998	1640	1142	683	351	250		
	RadStd	28	52	77	134	116	127	109	111	123	78	34	23		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page